Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/18/2009
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Pharmacokinetics & Contributing Physiologic Changes During Spaceflight - DSO 632B		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	lakshmi.putcha-1@nasa.gov, jason.boyd-1@nasa.gov	**Fax:**	FY 281-244-5734
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7760
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway		
PI Address 2:	Mail Code: SK		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	01/15/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Pre/Post Flight		
Flight Assignment:	ISS 13 NOTE: End date changed to 1/15/2009 (previously 6/1/2008) per HRP Master Task List information dtd 1/11/2012 (Ed., 2/23/2012) NOTE: Start/end dates changed to 5/15/2004-06/01/2008 (from 10/01/2005-9/30/2008) per JSC info rec'd 4/20/2009		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPERIMENT DESCRIPTION Lactulose and acetaminophen tests: Twenty-four hours before the test, participants are asked to avoid eating foods that are likely to generate hydrogen (such as cabbage, beans, dairy products). After approximate 8-hour fast (sleep period), participants will collect a saliva sample, a urine sample, and two consecutive breath samples, 5 minutes apart. They will then eat a low fiber breakfast, followed by ingestion of acetaminophen (650 mg.). After acetaminophen ingestion, participants will drink orange juice (or an agreed-upon fluid) containing 20 grams (g) of lactulose. Participants will not eat or exercise for 4 hours following ingestion. Drinking water is allowed. After drinking the acetaminophen, lactulose/orange juice mixture, participants will collect saliva and breath samples at the times shown in the table that follows, and will collect urine voids for 24 hours as separate samples. Participants will record intake of food, fluids, medications, and exercise during the 24-hour protocol periods. Glucose Breath Hydrogen test: After an overnight fast, a breath sample will be collected. Participants will then consume a dose of glucose dissolved in water. Breath samples will be collected every 15 minutes for 120 minutes post-dose. 13-C Urea Helicobacter test: After overnight fasting, a breath sample will be obtained. Participants will then consume a liquid test meal containing 13-C urea. Breath samples will be collected every 10 minutes for 30 minutes post-ingestion. OBJECTIVE: Determine changes in the GI function and physiology. Examine the pharmacokinetics of orally administered pharmaceutical probe acetaminophen. RELEVANCE Comparison of results from salivary pharmacokinetics of acetaminophen [DSO 458 “Pharmacokinetics of Acetaminophen” (7 subjects - Tylenol tablets) and DSO 622 “Gastrointestinal Function During Extended Duration Space Flight” (4 subjects - Tylenol syrup)] indicate absorption and bioavailability of oral liquid dosages may be better and more reliable than that of an equivalent solid dosage form. These factors may also improve in a fed state (ingesting a low fiber breakfast) than during fasted state (refraining from food for 8 - 10 hr.) as a result of compensating for the expected GI decrements during space flight. Unusually high breath levels of hydrogen (Shuttle data) and methane (Mir data), in addition to the reported regular use of countermeasure (lactobacillus) by the Russian program warrants the need for examining the microbial physiology of the GI tract. Decreased GI motility (as indicated from the limited results from the Extended Duration Orbiter Project (EDOMP) and the Shuttle/Mir Science Program (SMSP) DSO 622 “Gastrointestinal Function During Extended Duration Space Flight”) will affect drug absorption and bioavailability of oral medications and thus compromise therapeutic effectiveness during space flights. This effect will be compounded by anticipated changes in the hepatic metabolism of drugs and other xenobiotics. Estimating clinically significant organ functional changes and resultant pharmacokinetic consequences will lead to the design and development of effective pharmacologic countermeasures and therapeutic adjustments. See also https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004. [Ed. note: Task added to Task Book in March 2009]
Bibliography Type:	Description: (Last Updated: 06/23/2025)